**Code No: 111AH** **R13**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B.Tech I Year Examinations, September/October - 2021**
**ENGINEERING DRAWING**
**(Common to CSE, MIE, PTM)**

**Time: 3 Hours** **Max Marks: 75**

**Answer any three questions**
**All questions carry equal marks**
**- - -**

1.a) A circle of 50 mm diameter rolls inside another fixed circle of radius 80 mm. Draw the hypocycloid for the point P on the rolling circle, which is at the contact point of the fixed and rolling circles.

b) Construct a diagonal scale of RF = (1/4000) to show meters and long enough to measure 500 meters. Indicate on this a length of 374 meters. [12+13]

2. Draw the projections of a circle of 50 mm diameter when its plane is equally inclined to the HP and VP. One end of a diameter of the circle touches the HP while the other end touches the VP. [25]

3. A pentagonal prism of 30 mm base side and height 50 mm is standing on the HP on its base, with one side of base perpendicular to VP. This prism is cut by a section plane $40^0$ inclined to HP through mid-point of the axis. Draw the front view, sectional top view, and sectional side view. Draw also the true shape of the section. [25]

4. A cube of side 30 mm rests on its base in the HP with a vertical face inclined at $30^0$ to the VP. It is cut by a plane perpendicular to the VP and inclined at $50^0$ to the HP, and the top portion of cube is removed. The plane bisects the axis of the cube. Draw the development of the frustum of the cube. [25]

5. A rectangular prism 40 × 30 × 15 mm rests on the ground on one of its ends with one of the longest edges touching the PP and one of the shortest edges receding to the left at angle of $40^0$ to the PP. The nearest vertical edge is 15 mm to the left of the station point which is at a distance of 55 mm in front of the PP and 30 mm above the ground. Draw the perspective view of the prism. [25]

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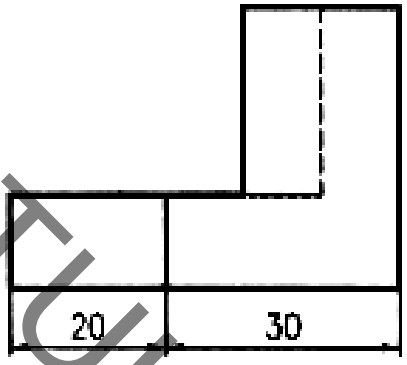
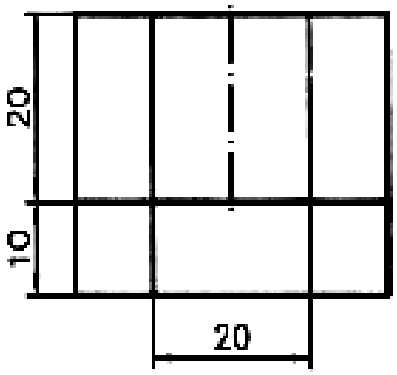
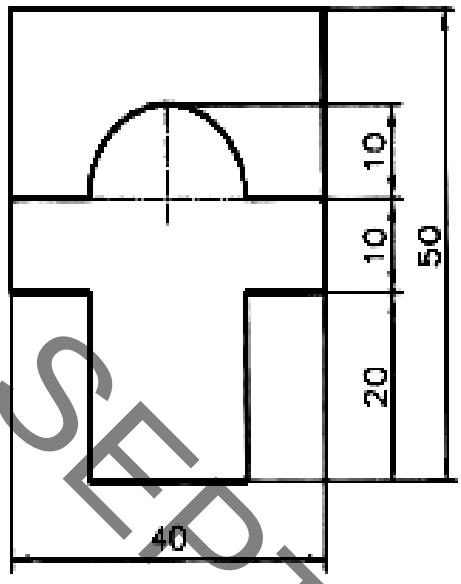
6. Three views of a solid are shown in the figure below. Draw the isometric projection of the solid. All dimensions are in mm. [25]




R.S. VIEW




FRONT VIEW




TOP VIEW

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